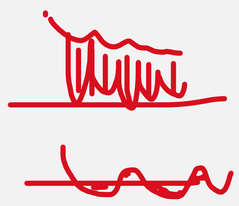
2024

3- $\int_{\pi}^{\infty} \frac{(4+\cos x)(x+2\sqrt{x})}{(e^{-x}+\sqrt{x})x}\,dx$



$(e^{-x}+\sqrt{x})x \sim x\cdot\sqrt{x}$

$+3 \leq 4+\cos x \leq 5$

$3\int \leq I \leq 5\int_{\pi}^{\infty} \frac{(x+2\sqrt{x})}{(e^{-x}+\sqrt{x})x}\,dx$

$\int_{\pi}^{\infty} \frac{1}{\sqrt{x}} + \frac{2}{x}\,dx \geq 2\int \frac{1}{x} = 2\ln(x)+C$

DV

$I \geq 3I'$ DV · I DV

$I = \int_0^1 \frac{e^{-x^2}+1}{e^{\sqrt{x}}-1}\,dx$ $\quad e^{\sqrt{x}}-1 \underset{x\to 0}{\to} 0$

$1 \geq e^{-x^2} \geq 0 \quad \forall x, -x^2 \leq 0, \exp((-\infty,0]) = (0,1]$

$2 \geq e^{x}+1 \geq 1$

$\int_0^1 \frac{1}{e^{\sqrt{x}}-1}\,dx \leq I \leq 2I'$

I'

$e^{y} = 1 + y + \frac{y^2}{2} + o(y^2)$

$= 1 + \sqrt{x} + \frac{x}{2} + o(x)$

$\int_0^1 \frac{dx}{\sqrt{x}+\frac{x}{2}+o(x)_{x\to 0}} = \int_0^1 \frac{1/2x \cdot dx}{1/\sqrt{x}+1+\frac{o(x)}{x}}$

$o(x) = O(x)$, $\int_0^1 1/\sqrt{x}\,dx = \left[\frac{1}{2}\sqrt{x}\right]_0^1 = \frac{1}{2}$

$\left(\int 1/\sqrt{x} = \frac{1}{2}\sqrt{x}\right)$

$e^{\sqrt{x}} \geq 1+\sqrt{x}$

$\int \frac{1}{e^{\sqrt{x}}-1}\,dx \leq \int_0^1 \frac{1}{\sqrt{x}}\,dx = 1/2$